FAISA ADITYA ATHALLAH

MACHINE LEARNING ENGINEER | AI NATIVE FULLSTACK DEVELOPER

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Informatics graduate from Universitas Amikom Yogyakarta specializing in Machine Learning and Fullstack Development. Driven by the challenge of bridging theoretical AI concepts with production-ready codebases. Possesses hands-on experience in engineering production-scale Retrieval-Augmented Generation (RAG) platforms and researching deep sequence recommendations using Transformer layers and Graph Neural Networks (GNNs). Leverages a strong visual execution background to streamline UI/UX collaboration, delivering intelligent systems that are both architecturally robust and highly user-focused.

PROJECTS & EXPERIENCE

AI Engineering / RAG · KnowledgeHub-AI | Next.js, Supabase, PostgreSQL, pgvector, Gemini AI, TypeScript

- Architected and deployed a production-scale Retrieval-Augmented Generation (RAG) platform to enhance context-aware information discovery.
- Engineered a serverless data pipeline for PDF parsing, text chunking, and multi-tenant vector embedding generation.
- Implemented secure tenant boundaries within a pgvector database to enable natural language querying with isolated semantic search.
- Integrated Google Gemini API for grounded reasoning, delivering natural language responses with verifiable source document citations.

Machine Learning Research · Dual View Contrastive Learning (Thesis) | PyTorch, Transformer, GNN, MovieLens-1M, Steam, LFM1b

- Designed a deep sequence recommendation network coupling Transformer self-attention blocks with Graph Neural Networks (GNN).
- Optimized representations to address dynamic user preferences and severe data sparsity issues using a self-supervised contrastive learning framework driven by InfoNCE loss.
- Benchmarked architectural outcomes and verified model performance using PyTorch across MovieLens-1M, Steam, and LFM1b datasets.

Frontend Engineering · Personal Portfolio Website | React, Vite, Tailwind CSS, Framer Motion, EmailJS

- Developed a modular React codebase hosted on Vercel incorporating smooth framer-motion reveals and a scroll spy navigation overlay.
- Structured a fully responsive CSS grid system via Tailwind CSS and integrated serverless email submission layers.

EDUCATION

Universitas Amikom Yogyakarta — **Sep 2022 - Jun 2026**

- Bachelor of Informatics (GPA: 3.19 / 4.00)
- Core Focus: Artificial Intelligence, Machine Learning, Data Mining, Software Engineering. Web Development

SKILLS

- Python, PyTorch, TensorFlow, Transformer Layers, Graph Neural Networks (GNN), Contrastive Learning, Retrieval-Augmented Generation (RAG), pgvector, Recommendation Systems.
- JavaScript, TypeScript, Next.js, React.js, Tailwind CSS, Supabase, PostgreSQL, SQL, REST API.
- Git, GitHub, Vercel, Google Gemini API, Serverless Pipelines.
- Graphic Design, Visual Asset Execution, Typography Configuration, Vector Pattern Engineering.
- Indonesian (Native), English.

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Lulusan Informatika dari Universitas Amikom Yogyakarta yang berspesialisasi dalam Machine Learning dan Pengembangan Fullstack. Terdorong oleh tantangan untuk menjembatani konsep teoretis AI dengan basis kode yang siap untuk tahap produksi (production-ready). Memiliki pengalaman langsung dalam merekayasa platform Retrieval-Augmented Generation (RAG) skala produksi dan meneliti rekomendasi sekuensial mendalam menggunakan lapisan Transformer dan Graph Neural Networks (GNN). Memanfaatkan latar belakang eksekusi visual yang kuat untuk menyederhanakan kolaborasi UI/UX, guna menghasilkan sistem cerdas yang kokoh secara arsitektur dan sangat berfokus pada pengguna.

PROYEK & PENGALAMAN

Rekayasa AI / RAG · KnowledgeHub-AI | Next.js, Supabase, PostgreSQL, pgvector, Gemini AI, TypeScript

- Merancang dan men-deploy platform Retrieval-Augmented Generation (RAG) skala produksi untuk meningkatkan penemuan informasi yang sadar konteks (context-aware).
- Merekayasa pipeline data nirserver (serverless) untuk penguraian (parsing) PDF, pemotongan teks (text chunking), dan pembuatan embedding vektor multi-tenant.
- Mengimplementasikan batasan tenant yang aman di dalam database pgvector untuk memungkinkan kueri bahasa alami dengan pencarian semantik yang terisolasi.
- Mengintegrasikan Google Gemini API untuk penalaran beralasan (grounded reasoning), memberikan respons bahasa alami dengan kutipan dokumen sumber yang dapat diverifikasi.

Riset Machine Learning · Dual View Contrastive Learning (Skripsi) | PyTorch, Transformer, GNN, MovieLens-1M, Steam, LFM1b

- Merancang jaringan rekomendasi urutan mendalam (deep sequence recommendation network) yang menggabungkan blok self-attention Transformer dengan Graph Neural Networks (GNN).
- Mengoptimalkan representasi untuk mengatasi preferensi pengguna yang dinamis dan masalah ketersebaran data (data sparsity) yang parah menggunakan kerangka kerja contrastive learning self-supervised yang didorong oleh InfoNCE loss.
- Melakukan uji tolok ukur (benchmark) terhadap hasil arsitektur dan memverifikasi kinerja model menggunakan PyTorch di seluruh dataset MovieLens-1M, Steam, dan LFM1b.

Rekayasa Frontend · Situs Web Portofolio Pribadi | React, Vite, Tailwind CSS, Framer Motion, EmailJS

- Mengembangkan basis kode React modular yang di-host di Vercel, menggabungkan efek kemunculan framer-motion yang halus dan overlay navigasi scroll spy.
- Menyusun sistem grid CSS yang sepenuhnya responsif melalui Tailwind CSS dan mengintegrasikan lapisan pengiriman email nirserver (serverless).

PENDIDIKAN

Universitas Amikom Yogyakarta — **Sep 2022 - Jun 2026**

- Sarjana Informatika (IPK: 3.19 / 4.00)
- Fokus Utama: Kecerdasan Buatan (Artificial Intelligence), Machine Learning, Penambangan Data (Data Mining), Rekayasa Perangkat Lunak, Pengembangan Web.

KEAHLIAN

- Python, PyTorch, TensorFlow, Lapisan Transformer (Transformer Layers), Graph Neural Networks (GNN), Contrastive Learning, Retrieval-Augmented Generation (RAG), pgvector, Sistem Rekomendasi (Recommendation Systems).
- JavaScript, TypeScript, Next.js, React.js, Tailwind CSS, Supabase, PostgreSQL, SQL, REST API.
- Infrastruktur & Alat: Git, GitHub, Vercel, Google Gemini API, Pipeline Nirserver (Serverless Pipelines).
- Desain: Desain Grafis, Eksekusi Aset Visual, Konfigurasi Tipografi, Rekayasa Pola Vektor.
- Bahasa: Bahasa Indonesia (Penutur Asli), Bahasa Inggris